

Beta Thetas

10/03/2026

Brandy Gonzales

We each worked independently. I uploaded the notebook to Collab and ran every cell. Logistic Regression scored 88.9% and the Decision Tree scored 83.3%. I tested other sets of three features. Proline, total phenols, and flavonoids scored 91.7%, and so did proline, total phenols and od280. Sets with malic acid scored as low as 66.7%. One choice I made was to keep proline and total phenols, because those two carried the improvement. My run lifted the score from 88.9% to 91.7%. If a teammate asked what the confusion matrix shows, I would say the diagonal is the correct guesses and the other squares are the misses.

Derrick K. Gibbs-McGlaston

I wrote my own reflection answers in the notebook, covering the three types of machines learning, my analysis of the Wine Classification results, a real-world application idea (phishing email detection in cybersecurity), and my key learnings. I also wrote a longer Module 3 reflection on the ML workflow, the model results, the challenges I faced, how this connects to my future work, and the questions I want to explore next.

Team Coordination

I communicated with my group by message about the assignment. I completed my own portion of the work individually and am only describing the work I personally did in this journal.

Marlon Roberts

For my contribution I ran through the analysis, after running the analysis I uploaded it on GitHub, from there I ran through the run test one more time to double check. After checking each cell to see if it was a complete test. Running the 25 code cells and troubleshooting any syntax or runtime

errors. Making sure the machine learning pipeline runs smoothly from data collection through model evaluation without breaking. I created the file and repository, everyone contributed separately and had good outcomes. After running through the cells, I write my reflection and provide what I contributed.

Pablo Lara

For this assignment, I completed my portion of the Module 3 Machine Learning lab. I worked through the Wine dataset and reviewed the different types of machine learning, including supervised, unsupervised, and reinforcement learning. I explored the dataset, checked the data for missing values, and reviewed the EDA visualizations.

I also worked through the machine learning workflow by selecting features, splitting the data into training and testing sets, training the Logistic Regression and Decision Tree models, and comparing their performance. I reviewed the accuracy results and confusion matrix to better understand how the models performed on data they had not seen before.

I completed the machine learning concepts assessment and my reflection and analysis. For my real-world application, I focused on using supervised machine learning in cybersecurity to help identify potentially malicious network activity.